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I/O UNIT AND BASE UNIT

Abstract

An I/O unit is arranged adjacent to another I/O unit along a first direction, and includes end engaging parts provided at an end section in a second direction orthogonal to the first direction; connection terminals provided on both sides of the first direction; and side engaging parts and side engaged parts provided respectively on one side and the other side of the first direction. The connection terminals are located at a position overlapping with the side engaging parts and the side engaged parts in the second direction, or on the end engaging parts side from the side engaging parts and side engaged parts.

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Background/Summary

TECHNICAL FIELD
[0001] The present invention relates to an I/O unit, and a base unit provided in the I/O unit.
BACKGROUND ART
[0002] JP 2018-056503 A discloses an I/O unit. According to JP 2018-056503 A, a plurality of I/O units are installed on a DIN rail.
SUMMARY OF THE INVENTION
[0003] The I/O units each include side terminals. The side terminals are provided on both sides of the I/O unit (see also JP 2018-056503 A). Two I/O units adjacent to each other on the DIN rail are electrically connected to each other by bringing their side terminals into contact with each other.
[0004] However, each I/O unit installed on the DIN rail may be tilted with respect to another adjacent I/O unit. As a result, the side terminals of the two adjacent I/O units are separated from each other.
[0005] The present invention has the object of solving the aforementioned problem.
[0006] According to a first aspect of the present invention, there is provided an I/O unit configured to be arranged adjacent to another I/O unit along a first direction, the I/O unit comprising: an end engagement portion provided at an end portion on one side in a second direction orthogonal to the first direction, the end engagement portion being configured to engage with an installation member extending in the first direction; connection terminals provided on both sides in the first direction, and electrically connected to another I/O unit arranged adjacent to the I/O unit; a side engaging portion provided on one side in the first direction and configured to engage with another I/O unit arranged adjacent to the I/O unit on the one side; and a side engaged portion provided on another side in the first direction and configured to engage with the side engaging portion of another I/O unit arranged adjacent to the I/O unit on the other side, wherein the connection terminals provided on the both sides in the first direction are provided at positions where the connection terminals overlap the side engaging portion and the side engaged portion in the second direction, or provided closer to the end engagement portion than are the side engaging portion and the side engaged portion in the second direction.
[0007] According to a second aspect of the present invention, there is provided a base unit to which a control module configured to perform signal processing is connected in an I/**0** unit configured to be arranged adjacent to another I/O unit along a first direction, the base unit comprising: an end engagement portion provided at an end portion on one side in a second direction orthogonal to the first direction, the end engagement portion being configured to engage with an installation member extending in the first direction; a mounting portion provided on another side in the second direction, the control module being inserted into the mounting portion along the second direction; connection terminals provided on both sides in the first direction, electrically connected to the control module inserted into the mounting portion, and electrically connected to another I/O unit arranged adjacent to the I/O unit; a side engaging portion provided on one side in the first direction and configured to engage with another I/O unit arranged adjacent to the I/O unit on the one side; and a side engaged portion provided on another side in the first direction and configured to engage with the side engaging portion of another I/O unit arranged adjacent to the I/O unit on the other

side, wherein the connection terminals provided on the both sides in the first direction are provided at positions where the connection terminals overlap the side engaging portion and the side engaged portion in the second direction, or provided closer to the end engagement portion than are the side engaging portion and the side engaged portion in the second direction.
[0008] According to the present invention, the possibility that the connection terminals of two adjacent I/O units are separated from each other is reduced.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0009] FIG. **1** is a perspective view of a communication system according to a reference example;
[0010] FIG. **2**A is a perspective view showing a front portion and a left side portion of an I/O unit according to the reference example, and FIG. **2**B is a perspective view showing a rear portion and a right side portion of the I/O unit according to the reference example;
[0011] FIG. **3** is a diagram schematically showing a state where the I/O unit is tilted with respect to an adjacent I/O unit in the reference example;
[0012] FIG. **4**A is a perspective view showing a front portion and a left side portion of an I/O unit according to an embodiment, FIG. **4**B is an exploded view of the I/O unit of FIG. **4**A, and FIG. **4**C is a perspective view showing a rear portion and a right side portion of the I/O unit of FIG. **4**A;
[0013] FIG. **5**A is a diagram schematically showing a plurality of I/O units arranged in a left-right direction, and FIG. **5**B is a diagram schematically showing a state where one of the plurality of I/O units of FIG. **5**A is tilted; and
[0014] FIG. **6** is a diagram showing an I/O unit according to a first modification.
DETAILED DESCRIPTION OF THE INVENTION
EMBODIMENT
[0015] FIG. **1** is a perspective view of a communication system **100** according to a reference example.
[0016] Note that a plurality of arrows shown in each drawing indicate respective directions (front, rear, left, right, up, and down) used in the following description. A left-right direction (a first direction), a front-rear direction (a second direction), and an up-down direction (a third direction) are orthogonal to each other. The left-right direction and the front-rear direction are parallel to a horizontal plane, for example.
[0017] The communication system **100** is a system for transmitting signals between a controller and a device **104**. The controller is, for example, a numerical controller that controls a machine tool. The device **104** is, for example, an actuator, a detector, or the like provided in the machine tool. The controller is not shown in the drawing. FIG. **1** shows only a part including an external terminal **104***a* of the device **104**.
[0018] The communication system **100** includes a plurality of I/O units (input/output units) **108**. The plurality of I/O units **108** are consecutively arranged on a predetermined installation member **110**.
[0019] The installation member **110** is, for example, a DIN rail. The extending direction of the installation member **110** illustrated in FIG. **1** is the left-right direction. That is, the installation member **110** illustrated in FIG. **1** is long in the left-right direction.
[0020] The plurality of I/O units **108** are attached to the front portion of the installation member **110**. In the illustration of FIG. **1**, the arrangement direction of the plurality of I/O units **108** coincides with the extending direction of the installation member **110**.
[0021] FIG. **2**A is a perspective view showing a front portion **108**F and a left side portion **108**L of the I/O unit **108** according to the reference example. FIG. **2**B is a perspective view showing a rear portion **108**B and a right side portion **108**R of the I/O unit **108** according to the reference example.

[0022] The I/O unit **108** includes an insertion-receiving portion **18**, a control module **20**, an end engagement portion **24**, and a plurality of connection terminals **26**.
[0023] The insertion-receiving portion **18** is provided in the front portion **108**F of the I/O unit **108**. The external terminal **104***a* can be connected to the insertion-receiving portion **18**.
[0024] The number of the insertion-receiving portions **18** provided in the I/O unit **108** is not particularly limited. Each I/O unit **108** illustrated in FIGS. **1**, **2**A, and the like is provide with a plurality of the insertion-receiving portions **18**. In this case, each illustrated I/O unit **108** may be connected to a plurality of the devices **104** according to the number of the provided insertion-receiving portions **18**.
[0025] The control module **20** is a module that performs necessary processing in the I/O unit **108**. The control module **20** is accommodated in the I/O unit **108**. The control module **20** includes, for example, a circuit board on which a predetermined electric circuit is formed. The electric circuit includes, for example, a control integrated circuit (IC).
[0026] The control module **20** is electrically connected to the external terminal **104***a* via the insertion-receiving portion **18**. Consequently, the control module **20** can communicate with the device **104** via the insertion-receiving portion **18**. In a case where a plurality of the devices **104** are connected to the I/O unit **108**, the control module **20** communicates with the plurality of devices **104**.
[0027] The end engagement portion **24** is provided at the rear portion (an end portion) **108**B of the I/O unit **108**. The end engagement portion **24** is provided to engage with the installation member **110**. For example, the end engagement portion **24** includes a claw that engages with a part of the installation member **110**. The I/O unit **108** is installed on the installation member **110** by the end engagement portion **24** engaging with the installation member **110**.
[0028] The plurality of connection terminals **26** are disposed on each of the left side portion **108**L and the right side portion **108**R of the I/O unit **108**. The plurality of connection terminals **26** are electrically connected to the control module **20**. Therefore, the device **104** is electrically connected to the plurality of connection terminals **26** via the control module **20**.
[0029] In the following description, the connection terminals **26** disposed on the left side portion **108**L are also referred to as connection terminals **26**L. Further, the connection terminals **26** disposed on the right side portion **108**R are also referred to as connection terminals **26**R.
[0030] In a case where two I/O units **108** are adjacent to each other in the left-right direction, the plurality of connection terminals **26**L of the I/O unit **108** disposed on the right side are connected to the plurality of connection terminals **26**R of the I/O unit **108** disposed on the left side. Consequently, the control module **20** of the I/O unit **108** disposed on the right side is communicably connected to the control module **20** of the I/O unit **108** disposed on the left side.
[0031] The two electrically connected control modules **20** can communicate with each other.
[0032] In this instance, according to the illustration of FIG. **1**, the plurality of I/O units **108** are consecutively arranged in the left-right direction. In this case, the control modules **20** of all the I/O units **108** are communicably connected to each other via the plurality of connection terminals **26** provided in each of the I/O units **108**.
[0033] The control module **20** of the I/O unit **108** disposed on the leftmost side or the rightmost side among the plurality of I/O units **108** is electrically connected to the controller described above. As a result, the controller and the device **104** are electrically connected to each other via at least one I/O unit **108**.
[0034] According to the communication system **100** described above, the controller can communicate with all the devices **104** connected to the I/O units **108**.
[0035] However, in the two I/O units **108** adjacent to each other in the left-right direction, if the plurality of connection terminals **26**L of the I/O unit **108** disposed on the right side and the plurality of connection terminals **26**R of the I/O unit **108** disposed on the left side are separated from each other, the above-described communication cannot be performed.

[0036] In view of this, it is preferable that the I/O unit **108** further includes a coupling mechanism for mechanically coupling the I/O unit **108** to another I/O unit **108** adjacent thereto in the left-right direction. Therefore, the I/O unit **108** further including a side engaging portion **28** and a side engaged portion **30** is considered.

[0037] The side engaging portion **28** and the side engaged portion **30** are disposed on different sides of the I/O unit **108**. For example, the side engaging portion **28** is disposed on the left side portion **108**L (FIG. **2**A). In this case, the side engaged portion **30** is disposed on the right side portion **108**R (FIG. **2**B).

[0038] The side engaging portion **28** includes, for example, a claw portion. On the other hand, the side engaged portion **30** includes, for example, a groove portion. In the two I/O units **108** adjacent to each other in the left-right direction, the side engaging portion **28** of the I/O unit **108** disposed on the right side engages with the side engaged portion **30** of the I/O unit **108** disposed on the left side.

[0039] Consequently, in the two I/O units **108** adjacent to each other in the left-right direction, it is expected that the plurality of connection terminals **26**L of the I/O unit **108** disposed on the right side and the plurality of connection terminals **26**R of the I/O unit **108** disposed on the left side can be prevented from being separated from each other.

[0040] Further, the side engaging portion **28** and the side engaged portion **30** serve to mechanically couple the I/O unit **108** to another I/O unit **108**, whereby it becomes unnecessary for each connection terminal **26** to have a structure for mechanical coupling to another I/O unit **108**. This can simplify the structure of each connection terminal **26**.

[0041] However, the I/O unit **108** described above has room for further improvement as described below. Specifically, in a case where the plurality of I/O units **108** are coupled to each other by using the side engaging portions **28** and the side engaged portions **30**, at least one I/O unit **108** may be tilted with respect to the adjacent I/O unit **108**. This may cause communication between the plurality of I/O units **108** to be cut off.

[0042] FIG. **3** is a diagram schematically showing a state where the I/O unit **108** is tilted with respect to the adjacent I/O unit **108** in the reference example.

[0043] For example, among the plurality of I/O units **108** illustrated in FIG. **3**, the I/O unit **108** disposed on the rightmost side is tilted with respect to the adjacent I/O unit **108**. Consequently, the connection terminals **26**L of the I/O unit **108** disposed on the rightmost side and the connection terminals **26**R of the adjacent I/O unit **108** are separated from each other. As a result, the I/O unit **108** disposed on the rightmost side cannot communicate with all the I/O units **108** disposed on the left side of this I/O unit **108**.

[0044] The reason why one of the two adjacent I/O units **108** tilts with respect to the other is that the engagement between the side engaging portion **28** and the side engaged portion **30** is loose. Specifically, even when the side engaging portion **28** and the side engaged portion **30** engage with each other, a gap is left between the side engaging portion **28** and the side engaged portion **30**. As a result, even after the side engaging portion **28** and the side engaged portion **30** have engaged with each other, one of the two adjacent I/O units **108** can be displaced with respect to the other. The tilting described above occurs as a result of one of the two adjacent I/O units **108** being displaced with respect to the other.

[0045] Several methods for preventing one of the two adjacent I/O units **108** from being tilted with respect to the other are considered. For example, eliminating the gap between the side engaging portion **28** and the side engaged portion **30** is considered as a first method. That is, it is considered to design the shapes of the side engaging portion **28** and the side engaged portion **30** so that no gap is formed when the side engaging portion **28** and the side engaged portion **30** engage with each other.

[0046] However, in a case where the first method is employed, there is a concern that the operation of engaging the side engaging portion **28** and the side engaged portion **30** will be more difficult than necessary. In addition, in a case where the first method is employed, the work of disengaging

the side engaging portion **28** and side engaged portion **30** that have been engaged also becomes difficult. Further, the first method is not practical because, at least for the side engaging portion **28** and the side engaged portion **30**, it does not allow any dimensional error that occurs during actual manufacturing or any wear during use. Therefore, the first method is not preferable as a solution for preventing one of the two adjacent I/O units **108** from being tilted with respect to the other.
[0047] Tightening together the two adjacent I/O units **108** using a coupling member such as a screw is considered as a second method. In this case, by the coupling member tightening the two I/O units **108** with a force as strong as possible, it may be possible to prevent one of the two adjacent I/O units **108** from being tilted with respect to the other.
[0048] However, in a case where the second method is employed, the operation of tightening together the plurality of I/O units **108** becomes a burden on an operator. The burden increases as the number of the I/O units **108** provided in the communication system **100** increases. Therefore, also the second method is not preferable in consideration of the labor of the operator.
[0049] Based on the above, an I/O unit **10** according to an embodiment will be described below. However, explanations overlapping with those of the above-described reference example will be omitted as much as possible in the following description. Elements that have been described in the reference example are denoted by the same reference numerals as in the reference example unless otherwise specified.
[0050] FIG. **4**A is a perspective view showing a front portion **10**F and a left side portion **10**L of the I/O unit **10** according to the embodiment. FIG. **4**B is an exploded view of the I/O unit **10** of FIG. **4**A. FIG. **4**C is a perspective view showing a rear portion **10**B and a right side portion **10**R of the I/O unit **10** of FIG. **4**A.
[0051] The I/O unit **10** is preferably divisible into a connector unit **12**, a communication processing unit **14**, and a base unit **16**, which will be described below.
[0052] The connector unit **12** is a part including the front portion **10**F of the I/O unit **10**. The connector unit **12** is disposed forward of the communication processing unit **14**.
[0053] The connector unit **12** includes an insertion-receiving portion **18**. The insertion-receiving portion **18** is disposed on the front portion **10**F. The number of the insertion-receiving portions **18** provided in the connector unit **12** is not particularly limited. The connector unit **12** illustrated in FIG. **4**A and the like includes a plurality of the insertion-receiving portions **18**.
[0054] The communication processing unit **14** includes a control module **20** and a housing **22**. The housing **22** is a casing that accommodates the control module **20**.
[0055] The connector unit **12** is coupled to the communication processing unit **14** from the front side of the communication processing unit **14** toward the rear side. In a case where the connector unit **12** and the communication processing unit **14** are coupled to each other, a device **104** connected to the insertion-receiving portion **18** of the connector unit **12** is electrically connected to the control module **20** of the communication processing unit **14**.
[0056] Therefore, in a case where the connector unit **12** and the communication processing unit **14** are coupled to each other, the device **104** connected to the insertion-receiving portion **18** of the connector unit **12** and the control module **20** of the communication processing unit **14** can communicate with each other.
[0057] The base unit **16** is a part including the rear portion (an end portion) **10**B of the I/O unit **10**. The base unit **16** is disposed rearward of the communication processing unit **14**. The base unit **16** includes an end engagement portion **24**, a plurality of connection terminals **26**, a side engaging portion **28**, a side engaged portion **30**, and a mounting portion **32**.
[0058] The end engagement portion **24** is provided at the rear portion **10**B. Since the I/O unit **10** includes the end engagement portion **24**, the I/O unit **10** can be installed on an installation member **110**.
[0059] However, in the present embodiment, the end engagement portion **24** may be omitted. That is, the plurality of I/O units **10** may be lined up without being installed on the installation member

110.

[0060] The plurality of connection terminals **26** are disposed on each of a left side portion **16**L and a right side portion **16**R of the base unit **16**. In the following description, the plurality of connection terminals **26** arranged on the left side portion **16**L are also referred to as connection terminals **26**L. Further, the connection terminals **26** disposed on the right side portion **16**R are also referred to as connection terminals **26**R.

[0061] Note that the left side portion **16**L of the base unit **16** is included in the left side portion **10**L of the I/O unit **10**. The right side portion **16**R of the base unit **16** is included in the right side portion **10**R of the I/O unit **10**.

[0062] A part of each of the plurality of connection terminals **26**L is exposed to the outside of the base unit **16** on the left side portion **16**L. Further, a part of each of the plurality of connection terminals **26**R is exposed to the outside of the base unit **16** on the right side portion **16**R. The plurality of connection terminals **26**L and the plurality of connection terminals **26**R are arranged substantially symmetrically in the left-right direction.

[0063] The plurality of connection terminals **26**L can be connected to the plurality of connection terminals **26**R provided in another I/O unit **10** installed on the left side of the I/O unit **10**. Similarly, the plurality of connection terminals **26**R can be connected to the plurality of connection terminals **26**L provided in another I/O unit **10** installed on the right side of the I/O unit **10**.

[0064] The mounting portion **32** is provided at a front portion of the base unit **16**. The mounting portion **32** is a portion of the base unit **16** where the communication processing unit **14** is mounted (inserted). The communication processing unit **14** is mounted on the mounting portion **32** from the front side of the base unit **16** toward the rear side, for example.

[0065] The communication processing unit **14** is mounted on the mounting portion **32**, whereby the communication processing unit **14** and the base unit **16** are coupled to each other. The plurality of connection terminals **26** are electrically connected to the control module **20** of the communication processing unit **14**.

[0066] The side engaging portion **28** is disposed on the left side portion **16**L. In this instance, the side engaging portion **28** is disposed at substantially the same position as the plurality of connection terminals **26**L or forward of the plurality of connection terminals **26**L in the front-rear direction. The substantially same position as the plurality of connection terminals **26**L in the front-rear direction is a position where the side engaging portion **28** overlaps the plurality of connection terminals **26**L when viewed in the up-down direction.

[0067] The side engaging portion **28** of the present embodiment is disposed at substantially the same position as the plurality of connection terminals **26**L in the front-rear direction (see also FIG. **5**A). In this case, the side engaging portion **28** is disposed at a different position from the plurality of connection terminals **26**L in the up-down direction.

[0068] The base unit **16** preferably includes a plurality of the side engaging portions **28** (**28**A, **28**B). The base unit **16** of the present embodiment includes a first side engaging portion **28**A and a second side engaging portion **28**B. The first side engaging portion **28**A is disposed above the plurality of connection terminals **26**L. The second side engaging portion **28**B is disposed below the plurality of connection terminals **26**L.

[0069] The side engaged portion **30** is disposed on the right side portion **16**R. In this instance, the side engaged portion **30** is disposed at substantially the same position as the plurality of connection terminals **26**R or forward of the plurality of connection terminals **26**R in the front-rear direction, so as to be engageable with the side engaging portion **28** of another I/O unit **10** installed on the right side of the I/O unit **10**. The substantially same position as the plurality of connection terminals **26**R in the front-rear direction is a position where the side engaged portion **30** overlaps the plurality of connection terminals **26**R when viewed in the up-down direction.

[0070] The side engaged portion **30** of the present embodiment is disposed at substantially the same position as the plurality of connection terminals **26**R in the front-rear direction. In this case,

the side engaged portion **30** is disposed at a different position from the plurality of connection terminals **26**R in the up-down direction (see also FIG. **4**C).
[0071] The number of the side engaged portions **30** provided on the base unit **16** is the same as the number of the side engaging portions **28** provided on the base unit **16**. For example, in the case of the present embodiment, the base unit **16** includes two side engaged portions **30** (**30**A, **30**B) (see also FIG. **4**C). A first side engaged portion **30**A is disposed above the plurality of connection terminals **26**R. A second side engaged portion **30**B is disposed below the plurality of connection terminals **26**R.
[0072] According to the present embodiment, in a case where one of two I/O units **10** adjacent to each other in the left-right direction is tilted with respect to the other, the possibility that the communication between the control modules **20** of the two I/O units **10** is cut off is reduced. The reason is as follows.
[0073] FIG. **5**A is a diagram schematically showing a plurality of the I/O units **10** arranged in the left-right direction.
[0074] The I/O unit **10** and other I/O units **10** are installed so as to be arranged in the left-right direction. The control modules **20** of two I/O units **10** adjacent to each other in the left-right direction are communicably connected to each other via the respective plurality of connection terminals **26**.
[0075] The number of the I/O units **10** to be arranged is not particularly limited. In a case where three or more I/O units **10** are arranged in the left-right direction, the control modules **20** of all the I/O units **10** are communicably connected to each other via the respective plurality of connection terminals **26**.
[0076] FIG. **5**B is a diagram schematically showing a state where one of the plurality of I/O units **10** of FIG. **5**A is tilted.
[0077] For example, in the illustration of FIG. **5**B, the plurality of I/O units **10** arranged in the left-right direction include an I/O unit **10**α disposed on the rightmost side, and an I/O unit **10**β disposed on the immediate left side of the I/O unit **10**α.
[0078] The I/O unit **10**α is titled with respect to the I/O unit **10**B. Consequently, a part of the left side portion of the I/O unit **10**α and a part of the right side portion of the I/O unit **10**β are separated from each other in the left-right direction.
[0079] However, the side engaging portions **28** of the I/O unit **10**α and the side engaged portions **30** of the I/O unit **10**β have been engaged with each other. In this case, the I/O unit **10**α is tilted about the side engaging portions **28** of the I/O unit **10**α and the side engaged portions **30** of the I/O unit **10**β.
[0080] Therefore, the side engaging portions **28** of the I/O unit **10**α are not largely separated in the right-left direction from the side engaged portions **30** of the I/O unit **10**β. In this case, also the plurality of connection terminals **26**L disposed at substantially the same position as the side engaging portions **28** of the I/O unit **10**α in the front-rear direction are not largely separated in the left-right direction from the plurality of connection terminals **26**R disposed at substantially the same position as the side engaged portions **30** of the I/O unit **10**β in the front-rear direction.
[0081] In this manner, according to the present embodiment, the plurality of connection terminals **26**L of the I/O unit **10**α and the plurality of connection terminals **26**R of the I/O unit **10**β are not separated from each other. Therefore, the communication between the control module **20** of the I/O unit **10**α and the control module **20** of the I/O unit **10**β is not cut off.
[0082] Further, according to the present embodiment, two side engaging portions **28** and two side engaged portions **30** are disposed so as to sandwich the plurality of connection terminals **26**L of the I/O unit **10**α and the plurality of connection terminals **26**R of the I/O unit **10**β. This restricts more strongly the separation between the plurality of connection terminals **26**L of the I/O unit **10**α and the plurality of connection terminals **26**R of the I/O unit **10**β.
[0083] Moreover, according to the present embodiment, the I/O unit **10** can be divided into a

plurality of units. Specifically, the I/O unit **10** can be divided into the connector unit **12** having a function of connecting to the device **104**, the communication processing unit **14** having a function of performing communication processing, and the base unit **16** having a function of connecting to another I/O unit **10**. As a result, the operator can perform maintenance on each of the connector unit **12**, the communication processing unit **14**, and the base unit **16**, for example.

Modifications

[0084] Modifications of the above-described embodiment will be described below. However, explanations overlapping with those of the above-described embodiment will be omitted as much as possible in the following description. Elements that have been described in the above-described embodiment are denoted by the same reference numerals as in the above-described embodiment unless otherwise specified.

(Modification 1)

[0085] FIG. **6** is a diagram showing an I/O unit **101** (**10**) according to a first modification. Note that this modification is based on the premise that the end engagement portion **24** is provided at the rear portion **10**B of the I/O unit **101**.

[0086] The side engaging portion **28** and the side engaged portion **30** may be disposed forward of the plurality of connection terminals **26**. The side engaging portion **28** or the side engaged portion **30** engages with another I/O unit **101**, thereby restricting the large separation in the left-right direction between the side engaging portion **28** or the side engaged portion **30**, and the end engagement portion **24** engaged with the installation member **110**.

[0087] On the other hand, the end engagement portion **24** engages with the installation member **110**, thereby restricting the large separation in the left-right direction between the end engagement portion **24** and the side engaging portion **28** or the side engaged portion **30** engaged with another I/O unit **101**.

[0088] As a result, the positional displacement in the left-right direction of the plurality of connection terminals **26** sandwiched between the side engaging portion **28** (the side engaged portion **30**) and the end engagement portion **24** in the front-rear direction is strongly restricted. This reduces the possibility that the connection terminals **26** of the two adjacent I/O units **101** are separated from each other.

(Modification 2)

[0089] The side engaged portion **30** may be provided on the left side portion **16**L. In this case, the side engaging portion **28** is provided on the right side portion **16**R.

[0090] Note that both the side engaging portion **28** and the side engaged portion **30** may be provided on each of the left side portion **16**L and the right side portion **16**R. For example, the side engaging portion **28** is provided above the plurality of connection terminals **26**L on the left side portion **16**L, and the side engaged portion **30** is provided below the plurality of connection terminals **26**L on the left side portion **16**L. In this case, the side engaged portion **30** is provided above the plurality of connection terminals **26**R on the right side portion **16**R, and the side engaging portion **28** is provided below the plurality of connection terminals **26**R on the right side portion **16**R.

(Modification 3)

[0091] The connection terminals **26**, the side engaging portion **28**, and the side engaged portion **30** may be provided in the communication processing unit **14** or the connector unit **12**.

(Combination of Plurality of Modifications)

[0092] The modifications described above may be appropriately combined within a range in which no contradiction occurs.

Inventions Obtained from Embodiment

[0093] Hereinafter, the inventions that can be grasped from the above-described embodiment and modifications will be described.

<First Aspect of Invention>

[0094] A first aspect of the present invention is the I/O unit (**10**) configured to be arranged adjacent to another I/O unit (**10**) along a first direction, the I/O unit including: the end engagement portion (**24**) provided at the end portion (**10**B) on one side in a second direction orthogonal to the first direction, the end engagement portion being configured to engage with the installation member (**110**) extending in the first direction; the connection terminals (**26**) provided on both sides in the first direction, and electrically connected to another I/O unit (**10**) arranged adjacent to the I/O unit; the side engaging portion (**28**) provided on one side in the first direction and configured to engage with another I/O unit (**10**) arranged adjacent to the I/O unit on the one side; and the side engaged portion (**30**) provided on another side in the first direction and configured to engage with the side engaging portion of another I/O unit (**10**) arranged adjacent to the I/O unit on the other side, wherein the connection terminals provided on the both sides in the first direction are provided at positions where the connection terminals overlap the side engaging portion and the side engaged portion in the second direction, or provided closer to the end engagement portion than are the side engaging portion and the side engaged portion in the second direction.
[0095] This reduces the possibility that the connection terminals of the two adjacent I/O units are separated from each other.
[0096] The side engaging portion may include the first side engaging portion (**28**A) and the second side engaging portion (**28**B) that are arranged so as to sandwich, in a third direction, the connection terminal provided on the one side in the first direction, the third direction being orthogonal to the first direction and the second direction, and the side engaged portion may include the first side engaged portion (**30**A) and the second side engaged portion (**30**B) that are arranged so as to sandwich, in the third direction, the connection terminal provided on the other side in the first direction. This further reduces the possibility that the connection terminals of the two adjacent I/O units are separated from each other.
<Second Aspect of Invention>
[0097] A second aspect of the present invention is the base unit (**16**) to which the control module (**20**) configured to perform signal processing is connected in the I/O unit (**10**) configured to be arranged adjacent to another I/O unit (**10**) along a first direction, the base unit including: the end engagement portion (**24**) provided at the end portion (**10**B) on one side in a second direction orthogonal to the first direction, the end engagement portion being configured to engage with the installation member (**110**) extending in the first direction; the mounting portion (**32**) provided on another side in the second direction, the control module being inserted into the mounting portion along the second direction; the connection terminals (**26**) provided on both sides in the first direction, electrically connected to the control module inserted into the mounting portion, and electrically connected to another I/O unit (**10**) arranged adjacent to the I/O unit; the side engaging portion (**28**) provided on one side in the first direction and configured to engage with another I/O unit (**10**) arranged adjacent to the I/O unit on the one side; and the side engaged portion (**30**) provided on another side in the first direction and configured to engage with the side engaging portion of another I/O unit (**10**) arranged adjacent to the I/O unit on the other side, wherein the connection terminals provided on the both sides in the first direction are provided at positions where the connection terminals overlap the side engaging portion and the side engaged portion in the second direction, or provided closer to the end engagement portion than are the side engaging portion and the side engaged portion in the second direction.
[0098] This reduces the possibility that the connection terminals of the two adjacent I/O units are separated from each other.
REFERENCE SIGNS LIST
[0099] **10**, **101**: I/O unit [0100] **16**: base unit [0101] **18**: insertion-receiving portion [0102] **20**: control module [0103] **24**: end engagement portion [0104] **26**, **26**L, **26**R: connection terminal [0105] **28**: side engaging portion [0106] **30**: side engaged portion [0107] **110**: installation member

Claims

1. An I/O unit configured to be arranged adjacent to another I/O unit along a first direction, the I/O unit comprising: an end engagement portion provided at an end portion on one side in a second direction orthogonal to the first direction, the end engagement portion being configured to engage with an installation member extending in the first direction; connection terminals provided on both sides in the first direction, and electrically connected to another I/O unit arranged adjacent to the I/O unit; a side engaging portion provided on one side in the first direction and configured to engage with another I/O unit arranged adjacent to the I/O unit on the one side; and a side engaged portion provided on another side in the first direction and configured to engage with the side engaging portion of another I/O unit arranged adjacent to the I/O unit on the other side, wherein the connection terminals provided on the both sides in the first direction are provided at positions where the connection terminals overlap the side engaging portion and the side engaged portion in the second direction, or provided closer to the end engagement portion than are the side engaging portion and the side engaged portion in the second direction.
2. The I/O unit according to claim 1, wherein the side engaging portion includes a first side engaging portion and a second side engaging portion that are arranged so as to sandwich, in a third direction, the connection terminal provided on the one side in the first direction, the third direction being orthogonal to the first direction and the second direction, and the side engaged portion includes a first side engaged portion and a second side engaged portion that are arranged so as to sandwich, in the third direction, the connection terminal provided on the other side in the first direction.
3. A base unit to which a control module configured to perform signal processing is connected in an I/O unit configured to be arranged adjacent to another I/O unit along a first direction, the base unit comprising: an end engagement portion provided at an end portion on one side in a second direction orthogonal to the first direction, the end engagement portion being configured to engage with an installation member extending in the first direction; a mounting portion provided on another side in the second direction, the control module being inserted into the mounting portion along the second direction; connection terminals provided on both sides in the first direction, electrically connected to the control module inserted into the mounting portion, and electrically connected to another I/O unit arranged adjacent to the I/O unit; a side engaging portion provided on one side in the first direction and configured to engage with another I/O unit arranged adjacent to the I/O unit on the one side; and a side engaged portion provided on another side in the first direction and configured to engage with the side engaging portion of another I/O unit arranged adjacent to the I/O unit on the other side, wherein the connection terminals provided on the both sides in the first direction are provided at positions where the connection terminals overlap the side engaging portion and the side engaged portion in the second direction, or provided closer to the end engagement portion than are the side engaging portion and the side engaged portion in the second direction.